

Article

Forecasting Forex Market Volatility Using Deep Learning Models and Complexity Measures

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Abstract: In this article, we examine whether incorporating complexity measures as features in deep learning (DL) algorithms enhances their accuracy in predicting forex market volatility. Our approach involved the gradual integration of complexity measures alongside traditional features to determine whether their inclusion would provide additional information that improved the model's predictive accuracy. For our analyses, we employed recurrent neural networks (RNNs), long short-term memory (LSTM), and gated recurrent units (GRUs) as DL model architectures, while using the Hurst exponent and fuzzy entropy as complexity measures. All analyses were conducted on intraday data from four highly liquid currency pairs, with volatility estimated using the Range-Based estimator. Our findings indicated that the inclusion of complexity measures as features significantly enhanced the accuracy of DL models in predicting volatility. In achieving this, we contribute to a relatively unexplored area of research, as this is the first instance of such an approach being applied to the prediction of forex market volatility. Additionally, we conducted a comparative analysis of the three models' performance, revealing that the LSTM and GRU models consistently demonstrated a superior accuracy. Finally, our findings also have practical implications, as they may assist risk managers and policymakers in forecasting volatility in the forex market.

Keywords: deep learning algorithms; complexity measures; recurrent neural networks; long short-term memory; gated recurrent units; hurst exponent; fuzzy entropy; econophysics; forex market; volatility



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1. Introduction

Market volatility has been the subject of a great deal of debate in recent decades, especially after the outbreak of the 2008 global financial crisis. Fluctuations within the financial market exert a direct influence on market expectations, thereby shaping investment decisions and the formulation of monetary and fiscal policies (Gong et al. 2024). In addition, financial market volatility plays a key role in many important fields of finance, such as portfolio allocation, derivative pricing, and financial risk management. Hence, the accurate estimation of market volatility is crucial for both investors and regulators.

Numerous methods have been proposed to estimate volatility in the literature. One of the main volatility estimators that has become increasingly popular in recent years is based on price ranges (Faldziński et al. 2024). Price ranges, defined as the differential between the highest and lowest logarithmic security prices within a specified time interval, have long been recognized in the field of finance (Feller 1951; Parkinson 1980; Garman and Klass 1980; Rogers and Satchell 1991; Yang and Zhang 2000; Alizadeh et al. 2002). Although the Range-Based volatility estimator is very simple to calculate, it provides valuable information about volatility. Moreover, volatility models based on it have been found to be more efficient than standard volatility models based on closing prices (Todorova 2011; Todorova and Husmann 2011; Faldziński et al. 2024).

Modeling and forecasting financial market volatility represent critical and complex challenges in modern financial research. Financial markets are inherently complex systems, with security prices being influenced by nonlinear interactions among heterogeneous agents and diverse external factors (Zitis et al. 2023a). This intrinsic complexity makes accurate modeling and forecasting particularly demanding. Nevertheless, significant progress has been made in the study of volatility over recent decades, marked by the development of a wide range of sophisticated econometric models (Dhingra et al. 2023). According to Xu and Ouenniche (2012), volatility forecasting models can be categorized into the following three primary groups: time series volatility models, implied volatility models, and hybrid models. Time series volatility models are further divided into the three following subcategories: historical volatility models, generalized autoregressive conditional heteroscedasticity (GARCH) models, and stochastic volatility (SV) models. However, despite their extensive use, traditional econometric models such as GARCH and SV have inherent limitations due to their parametric nature. They often fail to capture the nonlinear and highly complex dynamics of financial time series effectively. Moreover, while these models provide valuable insights into daily returns and volatilities, they are not well-suited for forecasting intraday volatility (Andersen and Bollerslev 1997). These limitations have motivated researchers from diverse fields, including artificial intelligence (AI) and complex systems science, to explore alternative approaches. Nowadays, machine learning (ML) methods, such as deep neural networks, have become an integral part of financial problem analysis. A recent review study, which collected 348 articles and reviews applying ML and AI methods in the field of finance from 2011 to 2021, observed an average growth rate of 34% in these publications (Ahmed et al. 2022). The authors also expect this rate to increase in the coming decades. The popularity of algorithms like artificial neural networks (ANNs) stems from their ability to function as generalized nonlinear forecasting models. Additionally, ANNs are nonparametric, data-driven techniques capable of capturing nonlinear data structures without requiring a priori assumptions regarding the underlying relationships in a given problem (Zhang et al. 2001). Therefore, the use of AI/ML techniques for modeling and forecasting financial time series, such as volatility, appears ideal. This hypothesis is further supported by a recent systematic literature review which reported that the efficacy of AI and ML techniques for volatility prediction is highly promising, often providing results that are comparable to or better than those of their econometric counterparts (Gunnarsson et al. 2024). However, it is essential to acknowledge that AI/ML models are not without limitations. A primary constraint lies in their dependence on big data to ensure their reliable application. This reliance can prove particularly challenging in certain contexts, notably in financial time series analysis, where the availability of sufficient data is not always guaranteed.

On the other hand, the field of complex systems is a relatively new and broadly interdisciplinary area that considers systems with many components. These systems could be social, physical, or biological. Given this diversity of systems, studying them within a single framework may seem unconventional. However, while most scientific disciplines focus on individual components, complex systems science emphasizes the relationships and interactions among the components within a system (Siegenfeld and Bar-Yam 2020; Zitis et al. 2021). Thus, seemingly different complex systems, such as ecosystems, human societies, the human brain, and financial markets, can be studied using similar mathematical measures. More specifically, the use of complexity measures to analyze financial markets is becoming increasingly widespread (Kutner et al. 2022). For example, measures such as entropy and the Hurst exponent are employed in finance because they provide valuable insights into aspects such as market volatility (Nikolova et al. 2020; Takaishi 2020; Jakimowicz 2020). Additionally, these measures are widely applied in evaluating the degree of market efficiency (Zitis et al. 2023b), as outlined in Fama's Efficient Market Hypothesis (Fama 1970).

Therefore, it is well-established that both measures derived from complex systems science (i.e., entropy and the Hurst exponent) and ML models are extensively used to

analyze financial markets. Given this, it is relevant to explore whether the integration of complexity measures and ML models can be effectively utilized in the analysis of complex systems, such as financial markets. A recent comprehensive literature review by [Raubitzek and Neubauer \(2021a\)](#) identified only 18 studies that combined ML and complexity measures for the analysis of real-world time series. For example, complexity measures have been employed to refine neural network architectures and algorithms ([Yakuwa et al. 2004](#); [Selvaratnam and Kirley 2006](#)), identify regions of increased predictability ([Ghosh et al. 2017](#); [Neto et al. 2018](#)), filter predictions or ensembles ([Raubitzek and Neubauer 2021b](#)), and for feature selection ([Ni et al. 2011](#)). Additionally, these measures have been incorporated as features within ML models. However, their application as features, particularly in the context of financial time series analysis, has received relatively limited attention. Specifically, [Karaca and Baleanu \(2020\)](#) applied three neural network architectures to predict seven stock market indices, incorporating wavelet entropy and the Hurst exponent, which improved prediction accuracy. In a related study, [Karaca et al. \(2020\)](#) used Support Vector Regression (SVR), Multi-Layer Regression (MLR), and Feed-Forward Back Propagation models to predict eight indices, finding that the Hurst exponent was crucial for accuracy, with the best results achieved when all three complexity measures (Shannon entropy, Rényi entropy, and Hurst exponent) were included. Similarly, [Kim et al. \(2020\)](#) enhanced US stock price direction predictions using effective transfer entropy (ETE) as a feature, with the LSTM and MLP models performing best. [Cho and Lee \(2022\)](#) developed a forecasting model integrating asymmetric fractality with DL models, demonstrating that asymmetric Hurst exponents effectively predict one-day-ahead returns, especially in volatile markets. Their two-stage forecasting model further showed a strong performance across varying volatility levels.

In this article, we investigate whether incorporating complexity measures as features in DL models can improve their accuracy in predicting forex market volatility. To the best of our knowledge, this is the first attempt to explore this approach for forecasting volatility in the world's largest market, namely the forex market. Specifically, we propose the development of volatility forecasting models that combine complexity measures and DL models to predict the intraday volatility of four highly liquid currency pairs. To estimate this volatility, we employed the Range-Based estimator, recognized as an information-rich proxy for true volatility ([Alizadeh et al. 2002](#); [Gallant et al. 1999](#)) and suitable for both daily and intraday data ([Vortelinos 2014](#)). As DL algorithms, we chose to use three specific neural network architectures, RNN, LSTM, and GRU, for two reasons. First, these architectures have been extensively utilized in the literature for predicting the volatility of financial markets ([Mashrur et al. 2020](#)). The second reason is that, according to [Gunnarsson et al. \(2024\)](#), who conducted an extensive literature review, ANNs incorporating memory, such as LSTM and GRU, were consistently found to rank among the top-performing models for predicting volatility. Also, as complexity measures, we applied the Hurst exponent and fuzzy entropy. Specifically, we analyzed the temporal evolution of these measures using overlapping sliding windows. These measures were selected because they are based on distinct fundamental principles and both provide valuable insights into time series data. The Hurst exponent, in particular, is recognized for its ability to reveal long-term dependencies or trends in time series ([Minadakis et al. 2012](#); [Zournatzidou and Floros 2023](#)). The presence of long-term dependencies indicates that the dynamics in the data are influenced by historical fluctuations over an extended period, resulting in consequential dependencies ([Lahmiri and Bekiros 2021](#)). In financial markets, there is a view that prices often exhibit trends, allowing past prices to be used, to some extent, in predicting future price changes. For example, when there is a common belief among the majority of market participants (in terms of trading volume) that the observed price of an asset is either overvalued or undervalued, market participants make trades towards the "correct" price, signaling the prevailing trend. In this context, the Hurst exponent is an effective measure for revealing such trends in financial time series. On the other hand, fuzzy entropy, and the broader concept of entropy, is fundamentally associated with quantifying the randomness of a time series. Randomness

is typically characterized by the absence of recognizable patterns. A financial time series is deemed somewhat predictable if it exhibits consistent price patterns. Conversely, it is considered to be entirely random if it lacks repetitive patterns, with participants selling or buying without any identifiable pattern (Delgado-Bonal 2019). In this context, entropy serves as a statistical measure of the level of randomness in a time series, based on the quantification of the presence and repetition of patterns (Zitis et al. 2023b).

In summary, the motivation for our study is threefold. First, we aim to illuminate a relatively underexplored area—the integration of complexity measures as features within DL models for the analysis of financial time series. By exploring this topic, we seek to contribute to the ongoing discourse and demonstrate that this approach has the potential to improve the accuracy of financial time series forecasting models. At this point, it is important to note that, in recent years, an increasing number of researchers have been exploring the integration of various feature types for financial time series forecasting. Notable examples of this include the application of sentiment analysis and image analysis in forecasting financial time series (e.g., Liu et al. 2017; Kirisci and Yolcu 2022). Second, we aim to examine whether DL models with more advanced architectures outperform simpler models in this context. Third, our findings can assist investors, risk managers, and market makers in forecasting volatility in the foreign exchange market.

2. Methods

This section briefly introduces the following three DL models: the RNN model (Section 2.1.1), the LSTM model (Section 2.1.2), and the GRU model (Section 2.1.3). Additionally, key concepts and formulas related to the Hurst exponent (Section 2.2.1) and fuzzy entropy (Section 2.2.2) are presented.

2.1. Deep Learning

Deep learning is a sophisticated method of ML that relies on ANNs. As a promising subset of ML (which, in turn, is a subset of AI), DL has garnered significant attention in recent years. Although it was developed in the field of computer science, its applications have expanded into diverse areas such as neuroscience, medicine, astronomy, and finance (Imrana et al. 2021). Compared to traditional ML techniques like random forests (RFs) and support vector machines (SVMs), DL offers advantages such as superior generalization capabilities and a robust training power for handling large datasets. Examples of DL approaches include deep neural networks (DNNs), convolutional neural networks (CNNs), deep belief networks (DBNs), and recurrent neural networks (RNNs).

2.1.1. RNN

Among the various DL approaches, RNNs have demonstrated promise for time series forecasting, including financial time series. The reason that RNNs can better handle time series is that they can store the feature information of historical data in their internal state (i.e., memory) (Liu et al. 2019). More specifically, RNNs are a type of neural network that uses previous outputs as inputs while retaining hidden layers (Berman et al. 2019; Kim and Kim 2015). Therefore, unlike traditional feed-forward networks, RNNs can remember what they have learned previously and make decisions based on this acquired knowledge (Bengio et al. 1994; Imrana et al. 2021). The following equation explains the function of the single RNN cell:

$$\begin{aligned} h_t &= \tanh(W[h_{t-1}, x_t] + b), \\ y_t &= h_t, \end{aligned} \quad (1)$$

where W is the weight matrix, b is the bias matrix, and h_t and h_{t-1} are hidden states at current the time step and previous time step, respectively. RNNs perform computations using the weights, biases, and activation functions for each element of the input sequence (Figure 1). Essentially a neuron in an RNN has a single hyperbolic tangent function in which h_{t-1} and x_t are combined and multiplied by some weight matrix, then a bias is added to it, followed by passing it through the hyperbolic tangent function, which gives

back h_t . The hyperbolic tangent function ($\tanh$) scales the values to fall within the range from -1 to $+1$.

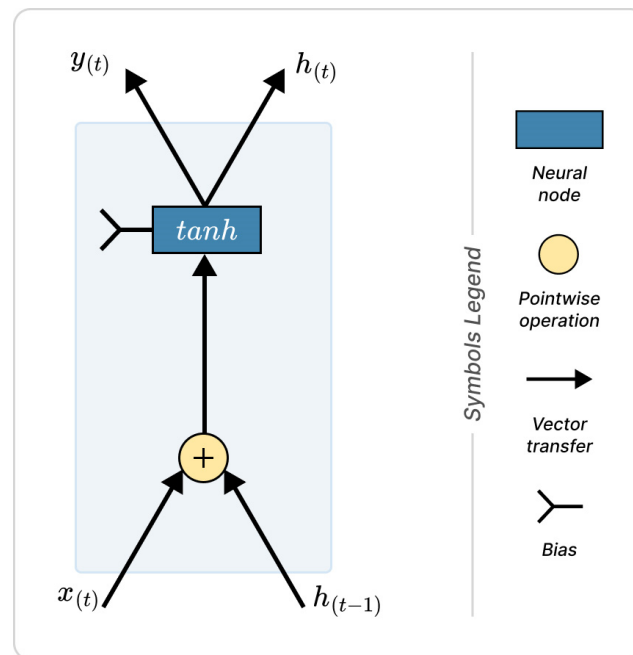


Figure 1. Schematic representation of simple RNN cell (Yu et al. 2019).

Although RNNs can address various research problems, they suffer from drawbacks such as vanishing gradients (Bengio et al. 1994; Pearlmutter 1995; Pineda 1987). This issue prevents them from effectively learning long-term dependencies, which means that the current position is influenced by the previous position. To overcome this limitation, Hochreiter and Schmidhuber (1997) proposed the LSTM network.

2.1.2. LSTM

Hochreiter and Schmidhuber (1997) proposed the LSTM network to address the shortcomings of traditional RNNs. In an LSTM cell, the memory is stored and converted from input to output in the cell state. An LSTM cell comprises the forget gate, input gate, update gate, and output gate. As their names suggest, the forget gate determines what to discard from previous memory units, the input gate decides what new information to accept into the neuron, the update gate updates the cell state, and the output gate generates the new long-term memory. These four main components work together uniquely, as LSTM processes long-term memory, short-term memory, and the input sequence at a given time step to generate new long-term memory, new short-term memory, and the output sequence at a given time step. The general architecture of an LSTM cell is shown in Figure 2. The input gate decides which information should be transferred to the cell and is mathematically represented as follows (ArunKumar et al. 2021):

$$i_t = \sigma(W_i * [h_{t-1}, x_t] + b_i). \quad (2)$$

The operator ‘ $*$ ’ denotes the element-wise multiplication of the vectors.

The information to be discarded from the previous memory is controlled by the forget gate, which is mathematically defined as follows:

$$f_t = \sigma(W_f * [h_{t-1}, x_t] + b_f). \quad (3)$$

The update gate is responsible for updating the cell state, which is expressed mathematically by the following equations:

$$\tilde{c}_t = \tanh(W_c * [h_{t-1}, x_t] + b_c), \quad (4)$$

$$c_t = f_t * c_{t-1} + i_t * \tilde{c}_t. \quad (5)$$

The output gate updates the hidden layer from the previous time step and is also responsible for updating the output, as given by the following:

$$o_t = \sigma(W_o * [h_{t-1}, x_t] + b_o), \quad (6)$$

$$h_t = o_t \tanh(c_t). \quad (7)$$

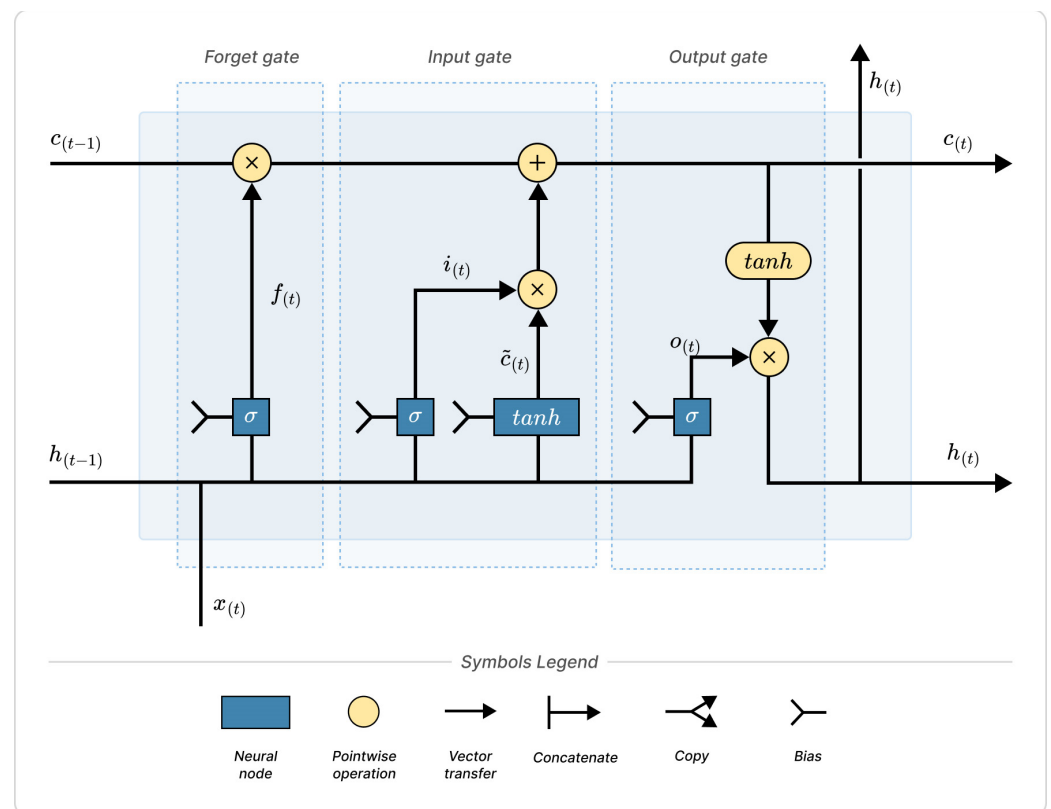


Figure 2. Schematic representation of LSTM cell (Yu et al. 2019).

2.1.3. GRU

Chung et al. (Chung et al. 2014) introduced GRU networks, which are a simplified, but equally popular, version of LSTM networks. The operation of a GRU cell is similar to that of an LSTM cell, but it uses a single hidden state that combines the forget gate and input gate into one update gate. Additionally, GRUs merge the cell state and hidden state into a single state, resulting in only two gates (update and reset gates) compared to the four gates in LSTM cells. This makes GRUs a simpler variant of LSTMs. The general architecture of a GRU cell is shown in Figure 3. The hidden state of the GRU cell is updated by the following equation (ArunKumar et al. 2021):

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t. \quad (8)$$

The update gate is computed using the following equation, which determines how much of the GRU unit is updated:

$$z_t = \sigma(W_z * [h_{t-1}, x_t]). \quad (9)$$

The reset gate is computed in a manner similar to the update gate, and is given by the following equation:

$$r_t = \sigma(W_r * [h_{t-1}, x_t]). \quad (10)$$

The new remember gate is generated by applying the hyperbolic tangent function to the reset gate, as described by the following equation:

$$\tilde{h}_t = \tanh(W * [r_t * h_{t-1}, x_t]). \quad (11)$$

Empirical studies have not definitively established which model is superior. However, GRUs require fewer network parameters, which makes them faster. In contrast, LSTM tends to provide a better performance when there are enough data and computational power (Alom et al. 2019).

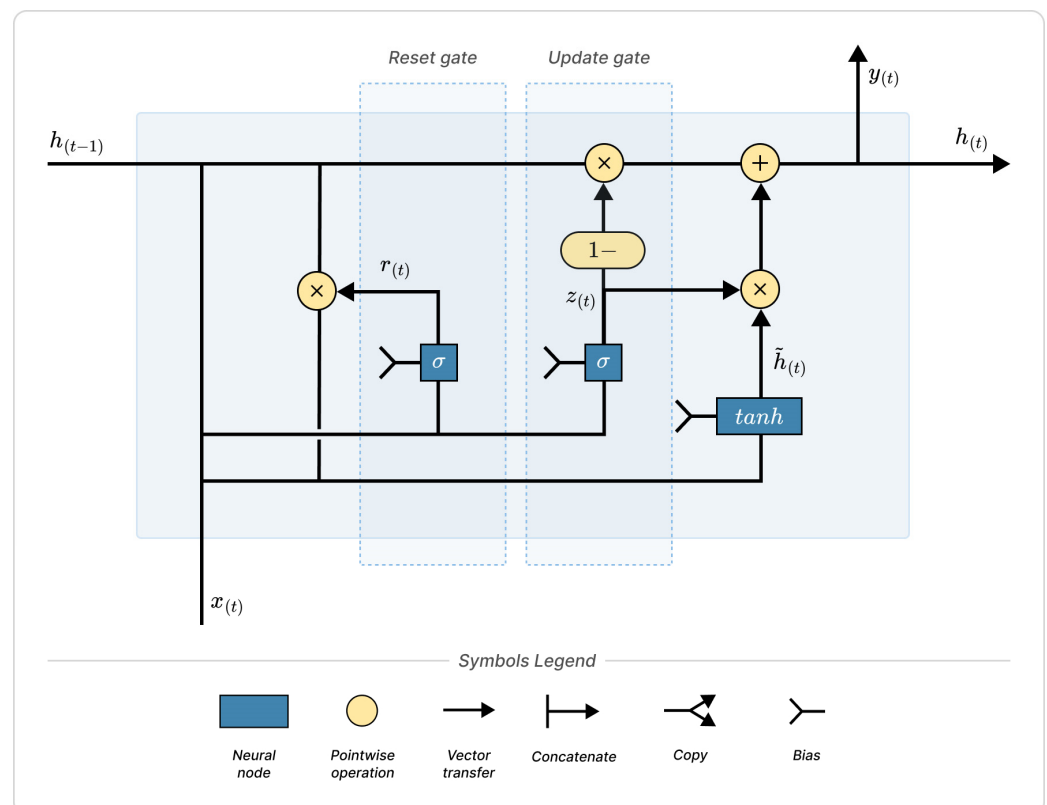


Figure 3. Schematic representation of gated recurrent unit (GRU) cell (Yu et al. 2019).

2.2. Complexity Measures

Numerous measures of complexity have been proposed in the literature. In this article, we select two widely recognized measures that have been extensively applied in financial time series analysis, as follows: the Hurst exponent and fuzzy entropy.

2.2.1. Hurst Exponent

The rescaled range analysis was developed by Hurst (1951) while investigating the statistical properties of Nile River overflows. This technique drew inspiration from Einstein's research on pure random walks, where the absolute value of particle displacement

scales in proportion to the square root of time (Couillard and Davison 2005). Hurst expressed the absolute displacement in terms of rescaled cumulative deviations from the mean and defined time as the number of data points “ n ” used. The scaling exponent of this relationship, now referred to as the Hurst exponent (H), provides information on the presence of long-range correlations in a time series. If the dataset is purely independent, the distance traveled increases with the square root of time, and the Hurst exponent will be $1/2$. If antipersistence exists in a system, the Hurst exponent will be less than $1/2$, and if persistence occurs, the Hurst exponent will be greater than $1/2$.

By calculating the rescaled range R/S_n values for different values of n , we can estimate the Hurst exponent H . We start with a time series of length N and divide it into M contiguous subperiods, each of length n , such that $M \times n = N$. Each subperiod is labeled I_m with $m = 1, 2, \dots, M$ and each element in I_m is labeled $N_{k,m}$ with $k = 1, 2, \dots, n$. For each subperiod I_m , the average and standard deviation are defined as follows:

$$\mu_m = \frac{1}{n} \sum_{k=1}^n N_{k,m}, \quad (12)$$

$$S_m = \sqrt{\frac{1}{n} \sum_{k=1}^n (N_{k,m} - \mu_m)^2}. \quad (13)$$

The second step is to construct a new time series for each subperiod I_m , where each series represents the accumulated departures from the mean value. This series is defined as follows:

$$X_{k,m} = \sum_{i=1}^k (N_{i,m} - \mu_m), \quad k = 1, 2, \dots, n \quad (14)$$

Note that $X_{n,m} = 0$. We study the range of these accumulated departures from the mean, as follows:

$$R_{I_m} = \max(X_{k,m}) - \min(X_{k,m}), \quad \text{where } 1 \leq k \leq n. \quad (15)$$

Each range is rescaled by dividing it by the standard deviation of its corresponding subperiod I_m . The final rescaled range value for length n is then defined as follows:

$$R/S_n = \frac{1}{M} \sum_{m=1}^M R_{I_m}/S_{I_m}. \quad (16)$$

The length n is then incremented to the next higher value. We only use values of n that encompass the first and last points of the time series to ensure that the same number of points is consistently used. Using $\log(R/S_n) = \log(c) + H\log(n)$, we run a regression to find the slope.

2.2.2. Fuzzy Entropy

Expanding upon the concepts established with approximate entropy (*ApEn*) and sample entropy (*SampEn*), Chen et al. (2007, 2009) combined elements from fuzzy sets and information theory to develop a fuzzy version of *SampEn*. Fuzzy entropy (*FuzzyEn*), like its predecessors *ApEn* and *SampEn* (Chen et al. 2009), is a “regularity statistic” that quantifies the (un)predictability of fluctuations in a time series. For estimating *FuzzyEn*, the similarity between vectors is defined based on fuzzy membership functions and the vectors’ shapes. The gradual and continuous boundaries of the fuzzy membership functions lead to several advantages, such as continuity, validity at small values, a higher accuracy, a stronger relative consistency, and less dependence on data length. *FuzzyEn* can be considered as an upgraded alternative to *SampEn* (and *ApEn*) for evaluating complexity, especially in short time series contaminated by noise (Balasis et al. 2013).

Similarly to *SampEn*, *FuzzyEn* excludes self-matches. However, it uses a slightly different approach for defining the initial $N - m$ vectors of a length of m , by removing a baseline, $\bar{s}_i$, as follows:

$$\bar{s}_i = m^{-1} \sum_{j=0}^{m-1} s_{i+j}, \quad (17)$$

i.e., for the *FuzzyEn* estimations, we use the first $N - m$ of the vectors, as follows:

$$\mathbf{X}_i^m = \{s_i, s_{i+1}, \dots, s_{i+m-1}\} - \bar{s}_i, \quad i = 1, 2, \dots, N - m + 1, \quad (18)$$

Then, the similarity degree, D_{ij}^m , between each pair of vectors, $\mathbf{X}_j^m$ and $\mathbf{X}_i^m$, being within a particular distance, r , from each other is defined by a fuzzy membership function, as follows:

$$D_{ij}^m = \mu(d_{ij}^m, r), \quad (19)$$

where d_{ij}^m is, as in the case of *ApEn* and *SampEn*, the supremum norm difference between $\mathbf{X}_i^m$ and $\mathbf{X}_j^m$. For each vector, $\mathbf{X}_i^m$, we estimate the average similarity degrees with respect to all other vectors, $\mathbf{X}_j^m$, $j = 1, 2, \dots, N - m + 1$, and $j \neq i$ (i.e., excluding itself), as follows:

$$\phi_i^m(r) = (N - m - 1)^{-1} \sum_{i=1, j \neq i}^{N-m} D_{ij}^m. \quad (20)$$

Then, we evaluate the following:

$$\varphi^m(r) = (N - m)^{-1} \sum_{i=1}^{N-m} \phi_i^m(r), \quad (21)$$

and the following:

$$\varphi^{m+1}(r) = (N - m)^{-1} \sum_{i=1}^{N-m} \phi_i^{m+1}(r). \quad (22)$$

The *FuzzyEn*(m, r) is then defined as follows:

$$FuzzyEn(m, r) = \lim_{N \rightarrow \infty} [\ln \varphi^m(r) - \ln \varphi^{m+1}(r)], \quad (23)$$

which, for finite time series, can be calculated by the following statistic:

$$FuzzyEn(m, r, N) = \ln \varphi^m(r) - \ln \varphi^{m+1}(r). \quad (24)$$

As mentioned earlier, *FuzzyEn* estimates the complexity of a dataset. Specifically, lower *FuzzyEn* values indicate a higher likelihood that a set of data will be followed by similar data, signifying a greater regularity. Conversely, higher *FuzzyEn* values suggest a lower likelihood of similar data being repeated, indicating more irregularity. Therefore, higher *FuzzyEn* values correspond to increased randomness, disorder, and system complexity, while lower values reflect a higher degree of order/organization, as well as a lower randomness and complexity.

3. Data, Model Optimization, and Evaluation

3.1. Data Description

The forex market is a decentralized market that operates 24 h a day, except on weekends. It is the world's largest financial market, characterized by a strong liquidity and transactions amounting to trillions of US dollars daily. In our article, we focused our analysis on four highly liquid currencies (the Euro (EUR), British pound (GBP), Canadian dollar (CAD), and Swiss franc (CHF)) against the US dollar (USD). For our analyses, we used intraday data (specifically, per 4 h intervals) during the period from 28 August 2014 to 29 De-

cember 2023. All financial time series were retrieved from Dukascopy Bank, a Swiss Forex Bank and an ECN broker with its headquarters in Geneva. The data are publicly available at <https://www.dukascopy.com/swiss/english/marketwatch/historical/> (last accessed 3 March 2024). In order to analyze the volatility of financial time series, we used one of the simplest and most efficient measures of volatility, known as “Range – Based Volatility”, which is defined as the difference between the highest and lowest logarithmic security prices over a fixed time interval, as follows:

$$(\text{Range} - \text{Based Volatility})_t = \log(\text{High}_t) - \log(\text{Low}_t). \quad (25)$$

Modifications of this volatility estimator have been used with great success in a wide range of analyses (e.g., Potirakis et al. 2013; Zitis et al. 2022), validating the views of Alizadeh et al. (2002) and Gallant et al. (1999), which support this volatility estimator as an information-rich proxy for true volatility. Furthermore, it is important to emphasize that employing Range-Based estimators for analyzing intraday data constitutes a robust methodological approach. Specifically, Andersen and Bollerslev (1998) highlighted that market microstructure issues, including non-synchronous trading effects, discrete price observations, and bid–ask spreads, can diminish the efficacy of intraday return variances or realized volatility as proxies or forecasts for volatility. In contrast, utilizing intraday high and low prices enhances the reliability and accuracy of volatility estimation (Chen et al. 2008).

3.2. Data Preprocessing

Data preprocessing is a crucial step in DL approaches and has a significant impact on the final results. One of the most commonly used data preprocessing techniques is data normalization. Nonlinear time series data, such as financial market data, often fluctuate across a large scale. Therefore, data normalization is essential for scaling the data to a smaller range, which, among other benefits, helps to accelerate the learning process of DL models. Numerous normalization techniques have been proposed in the literature. One of these techniques, which normalizes data to the range of [0, 1] using the following equation, is employed in our study:

$$x'_t = \frac{x_t - \min(x_t)}{\max(x_t) - \min(x_t)}, \quad (26)$$

where $\min(x_t)$ and $\max(x_t)$ are the minimum and maximum values of the time series, respectively. Given that the data are normalized during the model training phase, the test set output can be restored using the following formula:

$$x_t = x'_t(\max(x_t) - \min(x_t)) + \min(x_t). \quad (27)$$

3.3. Grid Search and Cross Validation Optimization

In ML models, parameters that require manual selection are known as hyperparameters. More specifically, hyperparameters are external parameters not inherent to the model itself and cannot be inferred from the dataset. Instead, they are configured by subject matter experts or through trial and error until an acceptable level of accuracy is achieved. For instance, an ANN has many such parameters, including the learning rate, the number of hidden layers, the number of nodes in each layer, and the optimizer. Hyperparameters significantly impact a model’s performance; incorrect selection can lead to underfitting or overfitting. One method for optimizing hyperparameters is Grid Search Cross-Validation (GSCV). This technique systematically explores all possible combinations of candidate hyperparameters, identifying the best-performing set. GSCV enhances the accuracy of the model’s generalization performance estimate. The grid search algorithm trains the model with each hyperparameter combination and selects the one with the lowest validation error. The algorithm’s search process is illustrated in Figure 4 (Alhakeem et al. 2022; Hong et al. 2022).

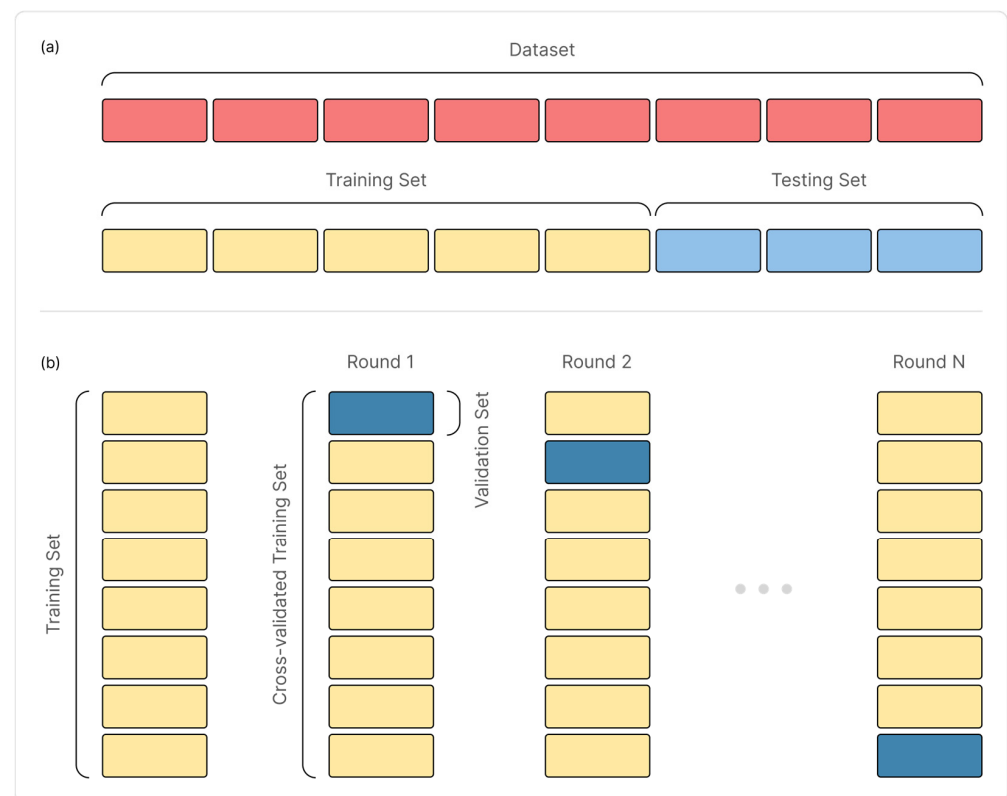


Figure 4. Partitioning of the dataset and the grid search process: (a) the division of the dataset into training and test sets and (b) a schematic representation of grid search and cross-validation.

3.4. Evaluation Measures

Evaluating the predictive performance of DL models is a crucial issue, as these measures form the basis for assessing and benchmarking the performance of each model. According to a recent review study (Kumbure et al. 2022), the majority of these measures can be classified into the following four categories: “statistical tests”, “return-based”, “accuracy-based”, and “error-based”. In our article, we utilized three of the most popular error-based metrics for measuring the forecasting performance of models (Mean Absolute Error—MAE, Root Mean Squared Error—RMSE, and Normalized Root Mean Squared Error—NRMSE) and an accuracy-based metric (Directional Symmetry—DS). The first metric that we utilized was MAE. Absolute-difference metrics like MAE limit the impact of individual outliers on model performance compared to the squared-difference metrics discussed below, making them particularly useful when dealing with data entry errors or other data quality issues (Steurer et al. 2021).

MAE measures the mean of the absolute differences between the observed values and predicted values, defined as follows:

$$MAE = \frac{1}{N} \sum_{t=1}^N |y_t - \hat{y}_t|, \quad (28)$$

where $\hat{y}_t$ is the forecasted value at time t , y_t is the observed value at time t , and N is the number of samples.

An alternative to mean absolute errors is mean squared errors. Squared-difference metrics are more sensitive to outliers than absolute-difference metrics, making them particularly useful in situations where minimizing large prediction errors is critical (Steurer et al. 2021). In our study, we selected RMSE and NRMSE. The difference between RMSE and NRMSE is that NRMSE is dimensionless, allowing for a comparison of values across different variables.

RMSE measures the root mean squared difference between the predicted and actual values, defined as follows:

$$RMSE = \sqrt{\frac{\sum_{t=1}^N (y_t - \hat{y}_t)^2}{N}}, \quad (29)$$

where $\hat{y}_t$ is the forecasted value at time t , y_t is the observed value at time t , and N is the number of samples.

NRMSE measures the normalized root mean squared difference between predicted and actual values, defined as follows:

$$NRMSE = \frac{1}{y_{max} - y_{min}} \sqrt{\frac{\sum_{t=1}^N (y_t - \hat{y}_t)^2}{N}}, \quad (30)$$

where $\hat{y}_t$ is the forecasted value at time t , y_t is the observed value at time t , and N is the number of samples.

The smaller the values of the three evaluation indicators, the more accurate the model's forecasting results.

On the other hand, *DS* measures the accuracy of a model by assessing its performance in predicting the direction of value changes, whether positive or negative. Specifically, the *DS* statistic represents the percentage of instances where the sign of the change in value from one time period to the next matches between the actual and predicted time series. A *DS* value of 100% indicates that the model perfectly predicts the direction of change in the time series from one period to the next (Sarveswararao et al. 2023).

DS is defined as follows:

$$DS = \frac{100}{N-1} \sum_{t=2}^N d_i, \quad (31)$$

$$d_i = \begin{cases} 1, & \text{if } (y_t - y_{t-1})(\hat{y}_t - \hat{y}_{t-1}) > 0 \\ 0, & \text{otherwise} \end{cases} \quad (32)$$

where $\hat{y}_t$ is the forecasted value at time t , y_t is the observed value at time t , N is the number of samples, and the direction of the value change is given by the corresponding d_i .

4. Results

As already mentioned, the forex market is a decentralized market that operates 24 h a day, except on weekends. It is the world's largest financial market, characterized by a strong liquidity and transactions amounting to trillions of US dollars daily. Its volatility significantly impacts imports and exports, capital flows, commodity prices, economic output, and employment. Therefore, the study of forex market volatility is a crucial issue. In this article, we study forex market volatility using DL models. Specifically, we investigate for the first time in the literature, to the best of our knowledge, whether incorporating complexity measures as features in DL models can improve their accuracy in predicting forex market volatility.

In our analysis, we first calculated the Range-Based Volatility estimator for four currency exchange rates, as outlined in Section 3.1. Subsequently, for the Range-Based Volatility time series corresponding to each currency rate, we estimated the Hurst exponent and *FuzzyEn* using sliding windows of 1024 samples. The dataset was then divided into training and test subsets, with 60% of the data allocated for training the DL models and 40% reserved for evaluating their performance (Figure 5).

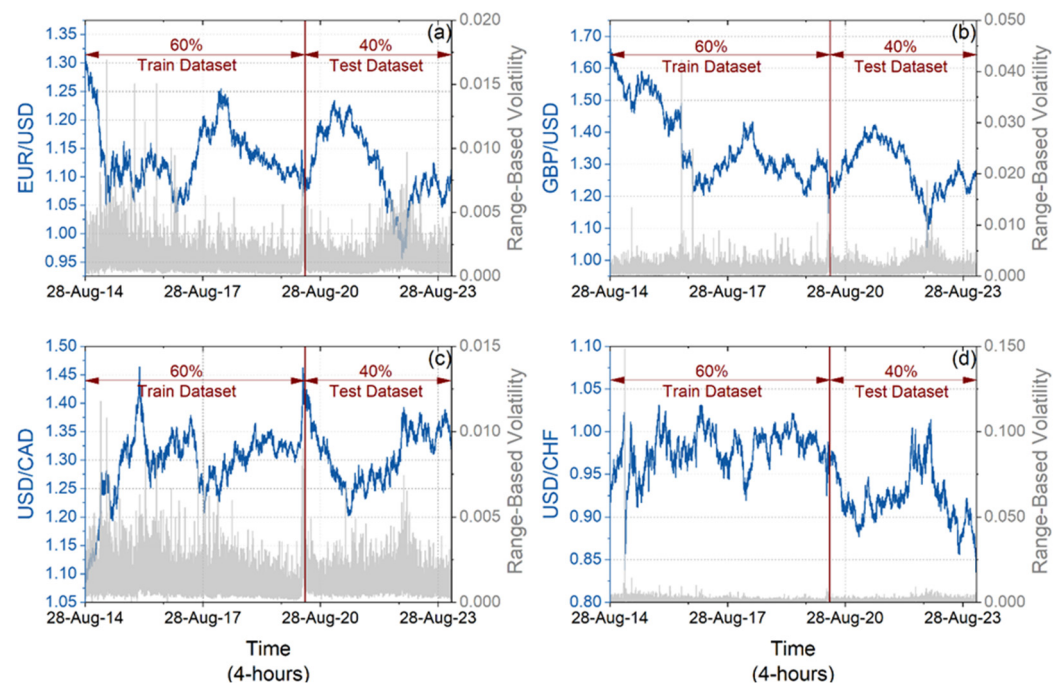


Figure 5. Evolution of four forex market currency exchange rate prices (blue curves, left vertical axis) and corresponding Range-Based volatility (grey curves, right vertical axis) over the period from 28 August 2014, to 29 December 2023: (a) EUR/USD, (b) GBP/USD, (c) USD/CAD, and (d) USD/CHF. The vertical red lines delineate the data area utilized for model training (on the left) and the data area employed for model testing (on the right).

For the training dataset, we applied normalization following the methodology described in Section 3.2. Thereafter, the DL architectures—namely RNN, LSTM, and GRU—were employed to independently forecast the Range-Based Volatility time series values for the subsequent 4 h period for each currency rate, utilizing the previous five samples of each variable as input memory. This procedure was repeated for each model using varying feature sets. Specifically, for each currency rate and model, the following feature combinations were used successively:

- Case I: Range-Based Volatility;
- Case II: Range-Based Volatility, High and Low;
- Case III: Range-Based Volatility, Hurst Exponent and *FuzzyEn*;
- Case IV: Range-Based Volatility, High, Low, Hurst Exponent, and *FuzzyEn*.

The main objective of this approach was to identify which combination of features enhanced the accuracy of the DL models. In particular, we aimed to investigate whether the inclusion of complexity measures, alongside traditional features, would provide additional insights and subsequently improve prediction accuracy. We concentrated on complexity measures, as they have been shown to offer significant insights into financial time series data (Jakimowicz 2020; Zitis et al. 2023a, 2023b; Tzouras et al. 2015). It is worth mentioning that this method of identifying optimal feature sets to improve model performance is widely recognized in the literature, e.g., (Cho and Lee 2022). It is also important to note that, for each case, we applied the GSCV methodology with three-fold cross-validation, as outlined in Section 3.3, to determine the optimal hyperparameters for each model. The list of hyperparameters and their respective search ranges were as follows: Activation: [Tanh], Batch size: [16, 32, 64], Epochs: [10, 50, 100], Optimizer: [Adam], and Neurons: [8, 16, 32, 64]. Additionally, we used a single hidden layer for each model, as adding more layers did not improve the results. The optimal hyperparameters for each model are summarized in Table 1.

Table 1. Optimal hyperparameter specifications for the RNN, LSTM, and GRU models.

Deep Learning Models	Hyperparameters	Optimal Hyperparameters															
		EUR/USD				GBP/USD				USD/CAD				USD/CHF			
		Case I	Case II	Case III	Case IV	Case I	Case II	Case III	Case IV	Case I	Case II	Case III	Case IV	Case I	Case II	Case III	Case IV
RNN	Activation	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh
	Batch size	16	16	16	64	16	16	16	16	16	16	16	16	32	16	16	32
	Epochs	10	50	50	50	50	100	10	50	10	10	50	100	10	100	100	100
	Optimizer	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam
LSTM	Neurons	32	16	32	16	64	32	64	64	16	16	8	32	32	32	32	64
	Activation	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh
	Batch size	16	16	16	16	16	16	16	16	16	16	16	32	16	16	16	16
	Epochs	10	100	100	100	100	50	100	50	100	50	100	100	50	50	100	100
GRU	Optimizer	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam
	Neurons	32	16	32	64	64	64	32	32	64	64	32	32	8	32	64	64
	Activation	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh	Tanh
	Batch size	16	16	16	32	16	16	16	16	16	16	16	16	16	32	32	16
	Epochs	10	100	50	50	10	100	100	100	50	100	50	100	100	100	100	100
	Optimizer	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam
	Neurons	32	16	32	64	64	64	32	32	64	64	32	32	8	32	64	64

After identifying the optimal hyperparameters, the models were trained, and predictions for the Range-Based Volatility time series were generated using the test dataset (Figures 6–9).

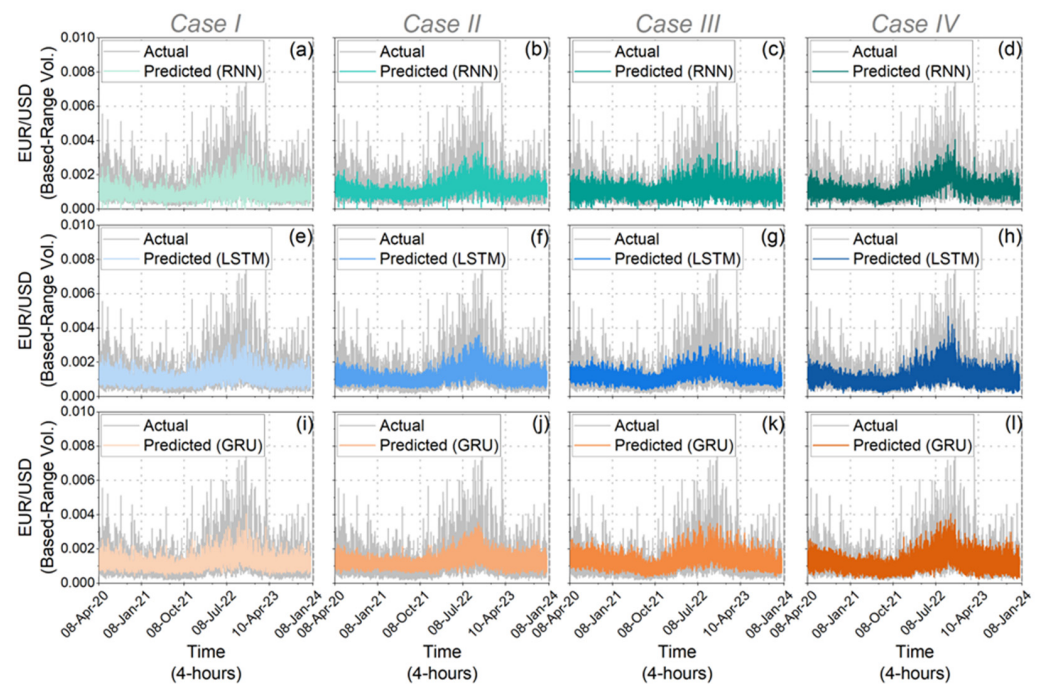


Figure 6. Actual values of the Range-Based volatility for EUR/USD (grey curves) and the model predictions (colored curves) by feature case and DL model for the test dataset (i.e., over the period 8 April 2020, to 29 December 2023). More specifically, subfigures (a,e,i) show the actual values of the Range-Based volatility and the predictions of the RNN, LSTM, and GRU models, respectively, for Case I (where the feature used was Range-Based Volatility). Subfigures (b,f,j) show the actual values of the Range-Based volatility and the predictions of the RNN, LSTM, and GRU models, respectively, for Case II (where the features used were Range-Based Volatility, High, and Low). Subfigures (c,g,k) show the actual values of the Range-Based volatility and the predictions of the RNN, LSTM, and GRU models, respectively, for Case III (where the features used were Range-Based Volatility, Hurst Exponent, and *FuzzyEn*). Subfigures (d,h,l) show the actual values of the Range-Based volatility and the predictions of the RNN, LSTM, and GRU models, respectively, for Case IV (where the features used were Range-Based Volatility, High, Low, Hurst Exponent, and *FuzzyEn*).

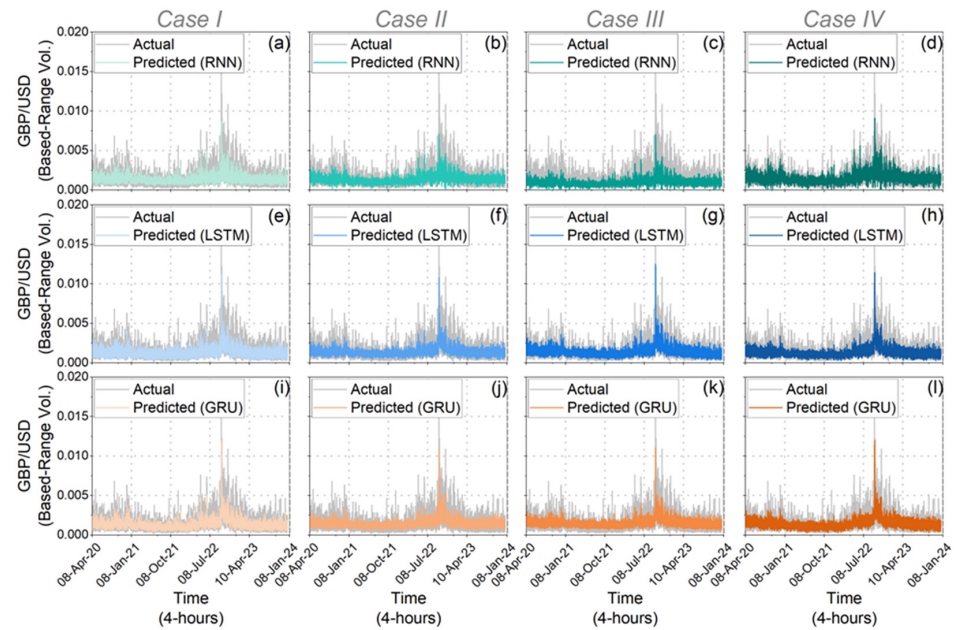


Figure 7. Actual values of the Range-Based volatility for GBP/USD (grey curves) and the model predictions (colored curves) by feature case and DL model for the test dataset (i.e., over the period 8 April 2020, to 29 December 2023). More specifically, subfigures (a,e,i) show the actual values of the Range-Based volatility and the predictions of the RNN, LSTM, and GRU models, respectively, for Case I (where the feature used was Range-Based Volatility). Subfigures (b,f,j) show the actual values of the Range-Based volatility and the predictions of the RNN, LSTM, and GRU models, respectively, for Case II (where the features used were Range-Based Volatility, High, and Low). Subfigures (c,g,k) show the actual values of the Range-Based volatility and the predictions of the RNN, LSTM, and GRU models, respectively, for Case III (where the features used were Range-Based Volatility, Hurst Exponent, and *FuzzyEn*). Subfigures (d,h,l) show the actual values of the Range-Based volatility and the predictions of the RNN, LSTM, and GRU models, respectively, for Case IV (where the features used were Range-Based Volatility, High, Low, Hurst Exponent, and *FuzzyEn*).

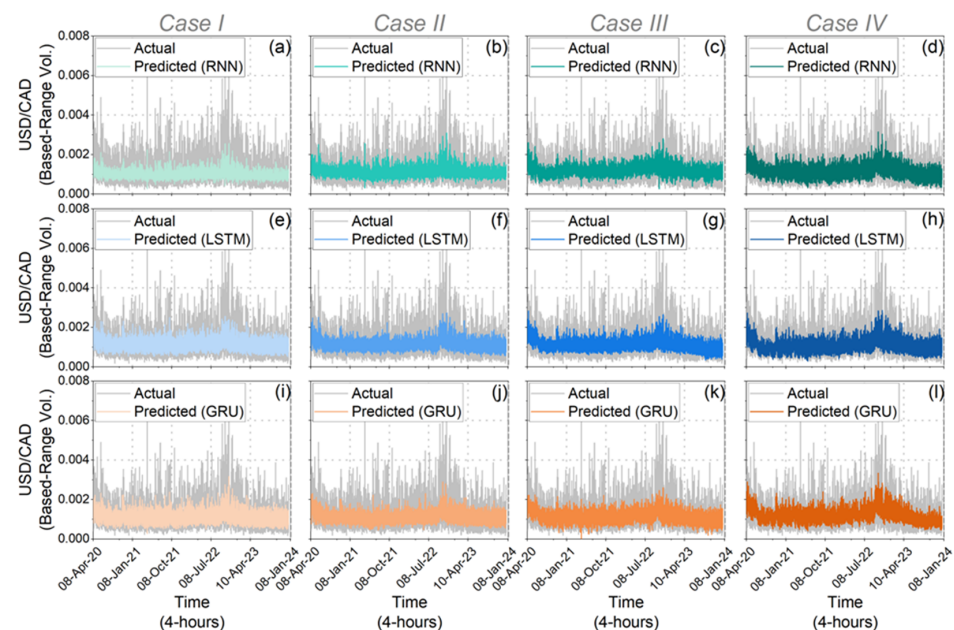


Figure 8. Actual values of the Range-Based volatility for USD/CAD (grey curves) and the model predictions (colored curves) by feature case and DL model for the test dataset (i.e., over the period 8

April 2020, to 29 December 2023). More specifically, subfigures (a,e,i) show the actual values of the Range-Based volatility and the predictions of the RNN, LSTM, and GRU models, respectively, for Case I (where the feature used was Range-Based Volatility). Subfigures (b,f,j) show the actual values of the Range-Based volatility and the predictions of the RNN, LSTM, and GRU models, respectively, for Case II (where the features used were Range-Based Volatility, High, and Low). Subfigures (c,g,k) show the actual values of the Range-Based volatility and the predictions of the RNN, LSTM, and GRU models, respectively, for Case III (where the features used were Range-Based Volatility, Hurst Exponent, and *FuzzyEn*). Subfigures (d,h,l) show the actual values of the Range-Based volatility and the predictions of the RNN, LSTM, and GRU models, respectively, for Case IV (where the features used were Range-Based Volatility, High, Low, Hurst Exponent, and *FuzzyEn*).

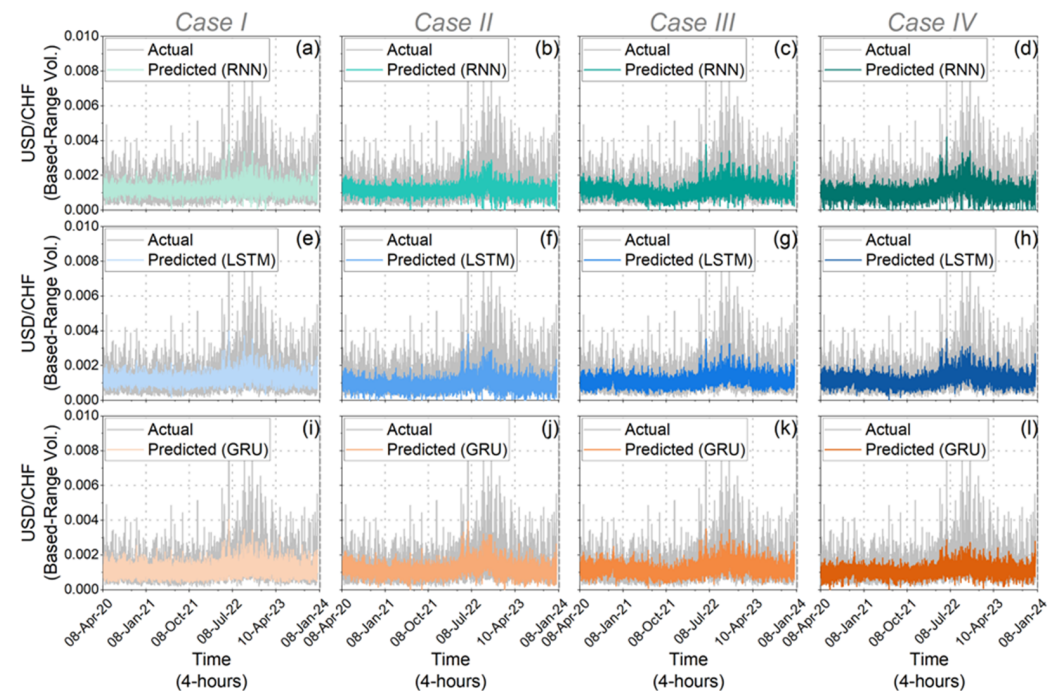


Figure 9. Actual values of the Range-Based volatility for USD/CHF (grey curves) and the model predictions (colored curves) by feature case and DL model for the test dataset (i.e., over the period 8 April 2020, to 29 December 2023). More specifically, subfigures (a,e,i) show the actual values of the Range-Based volatility and the predictions of the RNN, LSTM, and GRU models, respectively, for Case I (where the feature used was Range-Based Volatility). Subfigures (b,f,j) show the actual values of the Range-Based volatility and the predictions of the RNN, LSTM, and GRU models, respectively, for Case II (where the features used were Range-Based Volatility, High, and Low). Subfigures (c,g,k) show the actual values of the Range-Based volatility and the predictions of the RNN, LSTM, and GRU models, respectively, for Case III (where the features used were Range-Based Volatility, Hurst Exponent, and *FuzzyEn*). Subfigures (d,h,l) show the actual values of the Range-Based volatility and the predictions of the RNN, LSTM, and GRU models, respectively, for Case IV (where the features used were Range-Based Volatility, High, Low, Hurst Exponent, and *FuzzyEn*). A subsequent evaluation of the model performance on the test dataset was conducted using four statistical metrics (i.e., *MAE*, *RMSE*, *NRMSE*, and *DS*), as detailed in Section 3.4. This evaluation was performed for each of the three models, across all currency rates, and for each of the four feature sets. The results are presented in Table 2.

Table 2. Performance measures by currency pair, DL model, and feature case.

Deep Learning Models	Mean Absolute Error															
	EUR/USD				GBP/USD				USD/CAD				USD/CHF			
	Case I	Case II	Case III	Case IV	Case I	Case II	Case III	Case IV	Case I	Case II	Case III	Case IV	Case I	Case II	Case III	Case IV
RNN	0.00047	0.00048	0.00048	0.00047	0.00056	0.00055	0.00059	0.00058	0.00042	0.00043	0.00044	0.00041	0.00049	0.00048	0.00047	0.00047
LSTM	0.00047	0.00045	0.00047	0.00042	0.00052	0.00054	0.00052	0.00051	0.00041	0.00041	0.00040	0.00039	0.00049	0.00047	0.00045	0.00046
GRU	0.00049	0.00050	0.00049	0.00046	0.00055	0.00054	0.00056	0.00052	0.00040	0.00040	0.00041	0.00041	0.00046	0.00047	0.00046	0.00045
Deep Learning Models	Root Mean Squared Error															
	EUR/USD				GBP/USD				USD/CAD				USD/CHF			
	Case I	Case II	Case III	Case IV	Case I	Case II	Case III	Case IV	Case I	Case II	Case III	Case IV	Case I	Case II	Case III	Case IV
RNN	0.00074	0.00070	0.00071	0.00069	0.00087	0.00086	0.00096	0.00086	0.00062	0.00061	0.00060	0.00058	0.00069	0.00069	0.00069	0.00069
LSTM	0.00071	0.00066	0.00068	0.00065	0.00084	0.00082	0.00081	0.00080	0.00059	0.00059	0.00058	0.00058	0.00069	0.00073	0.00067	0.00066
GRU	0.00071	0.00068	0.00069	0.00066	0.00086	0.00080	0.00082	0.00079	0.00059	0.00058	0.00058	0.00057	0.00068	0.00068	0.00067	0.00068
Deep Learning Models	Normalized Root Mean Squared Error															
	EUR/USD				GBP/USD				USD/CAD				USD/CHF			
	Case I	Case II	Case III	Case IV	Case I	Case II	Case III	Case IV	Case I	Case II	Case III	Case IV	Case I	Case II	Case III	Case IV
RNN	1.84152	1.70613	1.75924	1.48703	1.65000	1.72154	2.10000	1.32000	2.86000	2.41000	2.22000	1.81922	2.20000	2.21000	1.88000	1.61000
LSTM	1.84268	1.51000	1.70091	1.36055	1.45000	1.52088	1.32076	1.37427	1.99000	2.20613	1.99000	1.76000	2.12000	1.94000	2.03000	1.80000
GRU	1.79000	1.63797	1.47042	1.25040	1.35000	1.43000	1.54000	1.24993	1.98000	2.16000	1.93000	1.66000	1.76000	1.70000	1.77000	2.02000
Deep Learning Models	Directional Symmetry															
	EUR/USD				GBP/USD				USD/CAD				USD/CHF			
	Case I	Case II	Case III	Case IV	Case I	Case II	Case III	Case IV	Case I	Case II	Case III	Case IV	Case I	Case II	Case III	Case IV
RNN	50.77%	52.92%	55.24%	54.79%	51.40%	53.13%	52.60%	53.40%	48.40%	49.00%	51.10%	55.41%	51.80%	53.50%	53.70%	57.20%
LSTM	51.72%	55.90%	55.02%	58.61%	55.10%	54.23%	57.32%	56.45%	55.00%	51.32%	55.20%	55.70%	52.10%	54.10%	54.80%	56.20%
GRU	51.30%	55.05%	55.08%	57.69%	50.80%	56.50%	57.10%	57.96%	55.80%	56.80%	54.40%	53.00%	55.60%	54.30%	55.00%	58.00%

Note: The feature cases with the highest performance for each model and exchange rate are highlighted using bold font.

The results indicate that, in the majority of cases, the LSTM and GRU models consistently outperform the simple RNN model. For instance, in Case IV for the EUR/USD exchange rate, a significant improvement in forecast accuracy is observed when comparing the GRU model to the RNN. Specifically, the improvements are approximately 2% for *MAE*, 4% for *RMSE*, 16% for *NRMSE*, and 5% for *DS*. Similar enhancements are noted in Case III for the GBP/USD exchange rate when comparing the LSTM model to the RNN, with improvements of approximately 12% for *MAE*, 16% for *RMSE*, 37% for *NRMSE*, and 9% for *DS*. Importantly, the enhanced predictive accuracy of the LSTM and GRU models is not limited to Cases III and IV, where complexity measures were included as features. For instance, in Case I for the USD/CAD exchange rate, the LSTM model demonstrates improvements of approximately 2% for *MAE*, 5% for *RMSE*, 30% for *NRMSE*, and 14% for *DS* when compared to the RNN. Similarly, in Case II for the USD/CHF exchange rate, the GRU model shows improvements of about 2% for *MAE*, 1% for *RMSE*, 23% for *NRMSE*, and 1% for *DS* over the RNN. This is a plausible finding, as both LSTM and GRU represent advancements over the basic RNN architecture. Furthermore, our findings align with those of previous research, which has demonstrated that the LSTM and GRU models exhibit a superior performance in forecasting the volatility of financial time series (Gunnarsson et al. 2024). Consequently, our results suggest that, for the analysis of currency rate volatility, models with more complex architectures, such as LSTM and GRU, are preferable to simpler models like the RNN.

The main finding of this study, aside from establishing that the LSTM and GRU models exhibit a higher predictive accuracy than the simpler RNN model, is that the accuracy of all models is consistently higher in Cases III and IV across various statistical metrics (Table 2). This suggests that incorporating complexity measures as features in DL models enhances their ability to predict volatility. Specifically, for the EUR/USD exchange rate, it is evident from all statistical measures that the accuracy of all three models improves when complexity measures are included as features (Table 2). For instance, a comparison of the LSTM model's performance in Case IV with Case I reveals improvements in prediction accuracy of approximately 11% for *MAE*, 8% for *RMSE*, 26% for *NRMSE*, and 13% for *DS*. The GRU model exhibits similar results; comparing Case IV with Case II, the improvements in accuracy are approximately 8% for *MAE*, 3% for *RMSE*, 24% for *NRMSE*, and 5% for

DS. Similarly, for the GBP/USD exchange rate, the LSTM model's performance in Case III compared to Case II shows improvements of approximately 4% for *MAE*, 1% for *RMSE*, 13% for *NRMSE*, and 6% for *DS*. Notably, even the simpler RNN model demonstrates an enhanced accuracy in predicting volatility when complexity measures are incorporated into its input. For example, in the case of the USD/CAD exchange rate, a comparison between Case IV and Case II reveals improvements in forecast accuracy of approximately 5% for *MAE*, 5% for *RMSE*, 25% for *NRMSE*, and 13% for *DS*.

In summary, the findings from our analyses suggest that the inclusion of complexity measures as features enhances the accuracy of volatility predictions. Conversely, when these measures are omitted, the models tend to perform less accurately, increasing the likelihood of underestimating or overestimating volatility. This can result in suboptimal decisions, such as inadequate risk management or inappropriate asset allocation, ultimately impacting returns and increasing exposure to unexpected losses. It is also noteworthy that, while the integration of complexity measures into DL models has not been extensively studied—and, to the best of our knowledge, no prior research has specifically applied these measures to DL models for predicting volatility in the forex market—our findings align with the broader literature. The studies we identified conclude that incorporating complexity measures as features in ML/DL models improves prediction accuracy (see Section 1: Introduction). Furthermore, our analyses reveal that, in most cases, the LSTM and GRU models outperform the RNN model in terms of volatility prediction accuracy. This result highlights the ability of LSTM and GRUs to efficiently capture long-term dependencies, unlike RNNs, where the current state is heavily influenced by the previous state.

5. Conclusions

In recent decades, predicting financial market volatility has become crucial for economic research, particularly for asset allocation and risk management. However, due to the nonlinear, chaotic nature of financial markets, traditional models often struggle to provide accurate forecasts. This challenge has spurred interest from AI and complex systems researchers, as machine learning models like ANNs can effectively capture nonlinearity without relying on prior assumptions, and complexity measures like the Hurst exponent and *FuzzyEn* reveal long-term dependencies and randomness. Nevertheless, despite the apparent suitability of integrating ML models with complexity measures for the analysis of financial time series, this area remains relatively underexplored.

In this article, we investigated whether the incorporation of complexity measures as features within DL models can enhance their accuracy in predicting volatility in the forex market, the largest financial market globally. Specifically, we proposed the development of volatility forecasting models that integrate the Hurst exponent and *FuzzyEn* as features into three DL architectures—RNN, LSTM, and GRU—to predict the intraday volatility of four highly liquid currency pairs (EUR/USD, GBP/USD, USD/CAD, and USD/CHF). To estimate volatility, we employed the Range-Based estimator, which is regarded as an information-rich proxy for true volatility. Our results demonstrated that the inclusion of complexity measures as features significantly enhanced the accuracy of volatility predictions. In contrast, when these measures were excluded, the models tended to perform less accurately, increasing the likelihood of underestimating or overestimating volatility, and consequently, risk. This result aligns with the literature, as the studies we identified conclude that incorporating complexity measures as features in DL models enhances prediction accuracy. Additionally, it was observed that models with more complex architectures, such as LSTM and GRUs, generally outperformed the simpler RNN model.

In summary, the contribution of our findings is threefold, as follows: (i) the primary conclusion is that incorporating complexity metrics as features in DL models can enhance model accuracy in predicting volatility. In achieving this, we illuminate a relatively underexplored area and aim to advance the discourse on the combined application of complexity metrics and DL models in finance; (ii) our results contribute to the existing literature by providing a comparative analysis of three popular DL architectures (RNN, LSTM, and GRU) in

terms of their accuracy in forecasting forex market volatility; and (iii) our results have practical applications, as accurate volatility predictions can help investors to enhance returns by signaling the optimal times to enter or exit markets, enabling them to capitalize on periods of high or low uncertainty. Furthermore, precise volatility forecasts allow risk managers to design effective hedging strategies to mitigate adverse price movements. Additionally, understanding volatility trends holds significant value for market makers, high-frequency trading (HFT) participants, and options traders. Accurate volatility forecasts enable these stakeholders to evaluate the likelihood of price fluctuations and effectively manage the risks associated with automated trading strategies.

Finally, as already mentioned, this scientific field remains relatively unexplored, offering considerable scope for future research. For instance, it would be valuable to examine the predictability of these models using alternative volatility estimators. Additionally, incorporating other complexity measures beyond the Hurst exponent and fuzzy entropy could provide further insights. Future research could also explore integrating complexity measures into DL models for directly predicting price, rather than focusing on volatility prediction. Moreover, applying these models to different time periods would be an interesting avenue for investigation.

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Conflicts of Interest: The authors declare no conflicts of interest.

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